

TEAL Electronics Corporation

Model DVMR MDP

TEAL Part Number 3600003

TABLE OF CONTENTS

SPECIFICATIONS	III
2 - PRODUCT DESCRIPTION	1
2-1 Product	1
2-2 Additional Features	1
3- PHYSICAL INSTALLATION	2
3-1 Unpacking	2
3-2 Mounting	2
4- CONNECTIONS	3
4-1 ABB Option	3
5- MAINTENANCE	4
5-1 FRU List	4
5-2 FRU Replacement Procedures	5
6- ENVIRONMENTAL CONDITIONS FOR TRANSPORT AND STORAGE	7
7- CONTACT INFORMATION	8
8- REVISION HISTORY	8
9- DRAWINGS	
Dimension Diagram	
Component Location (Internal)	
Schematic	
BVUPS Data sheet and schematic	

SPECIFICATIONS

COMMON SPECIFICATIONS	ALL MODEL NUMBERS
<i>Rated Operational ambient</i>	0° to 40°C
INDIVIDUAL SPECIFICATIONS	MODEL NUMBERS
<i>Model Number</i>	DVMR MDP
<i>(TEAL) Part Number</i>	3600003
<i>Overload Protection</i>	200 Amps
<i>Input Voltage Range</i>	380/400/415/480VAC +/- 10%
<i>Input Current</i>	151.5/144.9/138.9/120.3 Amps
<i>Output Voltages</i>	Output voltages is respective to input voltage
<i>Output Currents</i>	187 Amps Max at 380VAC
<i>Weight Pounds (Kg.)</i>	TBD

1- SAFETY



High voltage. Turn power off before servicing.



Electrical shock hazard. Lock out power at main panel. Failure to do so could result in serious injury or death.



To reduce the risk of electric shock, install in a temperature and humidity controlled indoor area relatively free of conductive contaminants.

2 - PRODUCT DESCRIPTION

2-1 Product

The Main Disconnect Panel (MDP) cabinet is designed to be used in the DVMR system to provide complete power distribution and Emergency-Off control for the Heat Exchanger Cabinet and the Power Distribution Unit (PDU). Power is provided directly from the facility through the Main Disconnect Panel to the entire MR product.

2-2 Additional Features

The MDP will provide two Remote Emergency-Off Switches with Emergency-Off labels for the customer to mount at accessible locations in the Magnet Room and Equipment Room. Emergency-Off switches are normally closed push buttons. The MDP will also provide two jumpers wire 58 and 59 and installed inside the MDP to allow it to be operated prior to the installation of the customer's remote Emergency-Off switches.



Jumper wires 58 and 59 must be removed immediately upon the installation of the remote Emergency-Off switches to enable their operation.

3- PHYSICAL INSTALLATION

3-1 Unpacking

- Protect finish, light, extended handles and buttons from damage while uncrating and placement.
- If forklift is used, lift only by frame, avoiding damage to sheet metal covers. If there is a removable door remove it while uncrating and during placement to avoid damage to hinges and latches.

If there is external damage or if internal components “rattle”, do not apply voltage. Check for any loose components or connections, call TEAL Tech Support for assistance.

All units are carefully and adequately packed. Please retain packing materials until the unit is satisfactorily install and accepted.

Damage in transit is always possible. Visible external damage should be noted on the bill of lading upon receipt and a claim filed with the shipper. Concealed damage can also be claimed with the shipper within 15 days. Contact TEAL if you wish assistance in filing your claim.

Each unit is inspected and tested before it leaves the factory. An individual check-out sheet is maintained on file by TEAL and retrievable by power size and serial number. Items tested include voltages input/output, current, and spike attenuation both normal and common mode. All connections are tight when they leave the factory.

3-2 Mounting

Refer to the attached mechanical drawing for dimensional information. The unit should be mounted in a dry location. The recommended upper limit installation height is 1.7m.

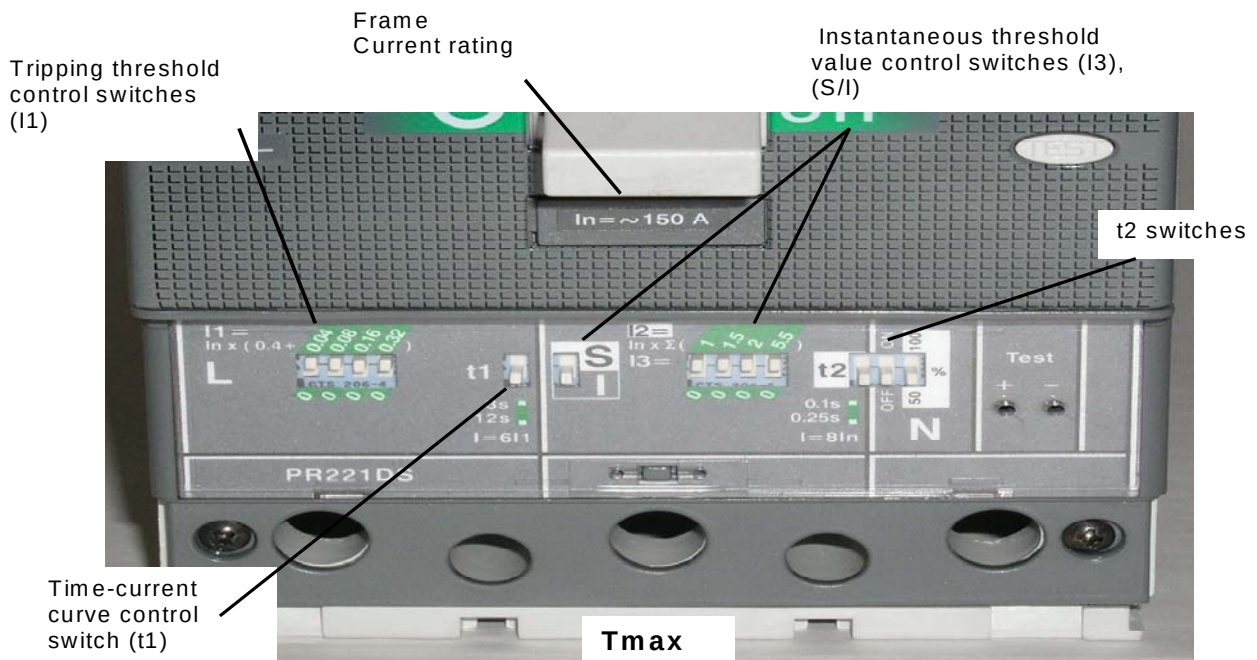


HIGH VOLTAGE. Turn power off before removing cover.

4- CONNECTIONS

Main Input connection will be to top of CB1. PDU Output will be connected to bottom of CB2 and HEC Output will be connected to top of CON1. Call TEAL Tech Support if you need advice on these connections.

4-1 ABB Option



Three dip-switches on the circuit breaker panel control the input voltage overload setting, the overload delay, and the short-circuit trip adjustment. Factory settings for the three switches are:

- I1 - set based on the input voltage. Refer to the Schematic for a table of the other I1 settings.
- t1 - longest delay (set to 12s)
- I/S – longest delay (set to I position)
- t2 – All three dip switches in down position
- I3 – All four dip switches in up position

Use a small blade screwdriver to move the dip switch toggles to the appropriate settings.

5- MAINTENANCE

It is recommended that the MDP battery be replaced every five years. Other than the battery, no maintenance is required other than periodically (i.e. every 6 months) all electrical connections should be checked for tightness. Any accumulations of dust or debris should be removed to keep ventilation working for convection cooling.

5-1 FRU List

Circuit diagrams, component part lists, descriptions, calibration instructions, or other information is available upon request from Teal Electronics for the purpose of repair or replacement. Refer to contact information in section 7.

Ref. Designator	Description	Part number
AUX2	CONTACTOR, AUX CONTACT, 1N.O./N.C. ABB #CAL5-11	0410155
BATT	BATTERY, RECHARGEABLE LEAD ACID, 12V, 7AH POWERSONIC #PS-1270	0870165
BVUPS	RELAY, BATTERY VOLTAGE UPS, 12V, TRANSTRONICS #BVUPS12FPA-10	0410288
CB1	CB, 250.0A, 3P, 600V ABB #T4N250BW (TMAX)	0260405
CB2	CB, 150.0A, 3P, 600V, ABB #S3N150TW	0260273
CB3	CB, 50.0A, 3P, 480V ABB #T1N050TL (TMAX)	0450619
CB4	CB, 3.0A, 1P, 277/480YABB#S201-K3	0450674
CB5	CB, 2A, 1P, 500 VDC ABB #S281UC-K2	0450835
CON1	CONTACTOR, 65.0A, 120VAC 3-POLE, ABB #A63-30-11-84	0410287
F1-F2	CB, FUSE, 1.5A, 600VAC 13/32" X 1 1/2" BUSSMAN #FNQ-R-1 1/2	0450836
F3-F5	CB, FUSE, 60.0A, 600VAC/ 500VDC, 100Ka IR SHAWMUT #AJT60	0450833
F6-F7	CB, FUSE, 1.0A, 500V AC/DC FAST ACTING BUSSMAN #KLM-1	0450545
IND1	LIGHT, GREEN UNIT CASE, 120VAC ABB#CBK-KLFL1G	0690079
PS1	POWER SUPPLY, 15W, 12VDC OUTPUT, 100-240VAC INPUT PULS #ML15.121	0870166
RY1-RY2	RELAY, 10A, 12VDC, DPDT IDEC #RR2BA- UDC12V	0410290
SW1	SWITCH, N.C. BLOCK FOR CH PB's CUTLER HAMMER #E22B1	0600024
SW2	SWITCH, N.O. BLOCK FOR CH PB's CUTLER HAMMER #E22B2	0600022
T1	XFORMER, CNTRL, 300VA 208/240/380/416/480\120VSQD #9070T300D40	0710100

5-2 FRU Replacement Procedures

⚠ DANGER



High voltage. Turn power off before servicing.



⚠ DANGER

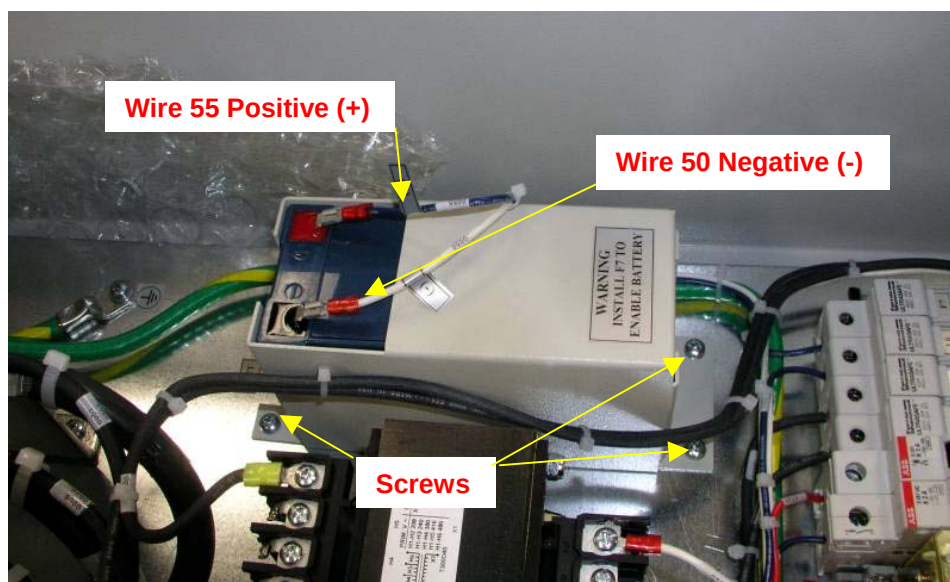


Electrical shock hazard. Lock out power at main panel. Failure to do so could result in serious injury or death.

⚠ IMPORTANT

Schematics should be used when replacing parts for wiring purposes.

- **Battery, PowerSonic # PS-1270, TEAL P/N 0870165**
Reference Designator: BATT
Tools: Philips head screwdriver
Size of Mounting Hardware: 10x32 Screws
1. Open front door on panel
 2. Disconnect wires 50 & 55 from the battery
 3. Remove the 4 screws securing the battery cover to the component plate



4. Remove the old battery

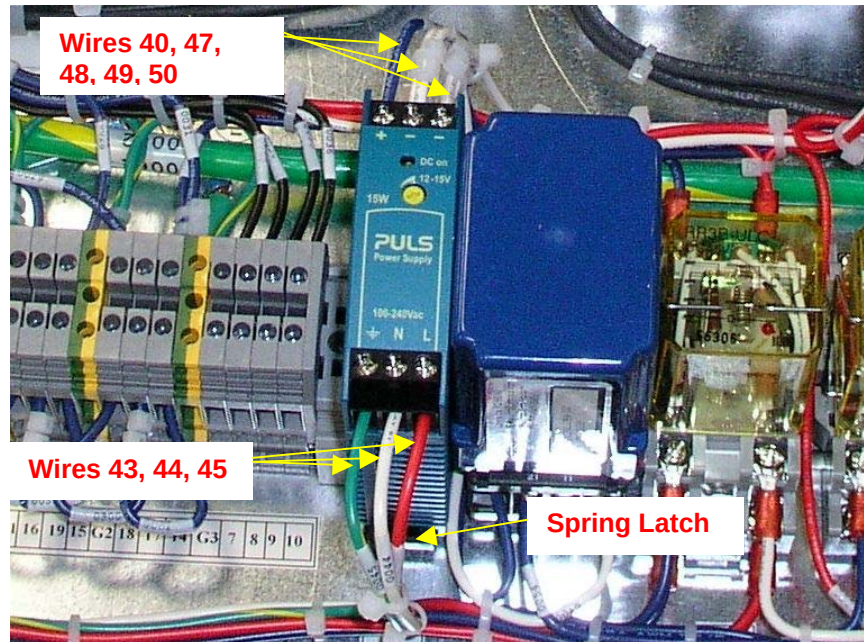
5. Installation is the reverse of removal. It is important to make sure the positive wire (55) goes to the + terminal on the battery

- **Power Supply, PULSPower # ML15.121, TEAL P/N 0870166**

Reference Designator: PS1

Tools: Philips head screwdriver, Flat head screwdriver

1. Open front door on panel



2. Disconnect wires 40, 43, 44, 45, 47, 48, 49 & 50 from the power supply
3. Use a flat head screw driver to release the spring clip holding the power supply to the DIN rail
4. Installation is the reverse of removal

- **UPS, Transtronics # BVUPS12PFA-10, TEAL P/N 0410288**

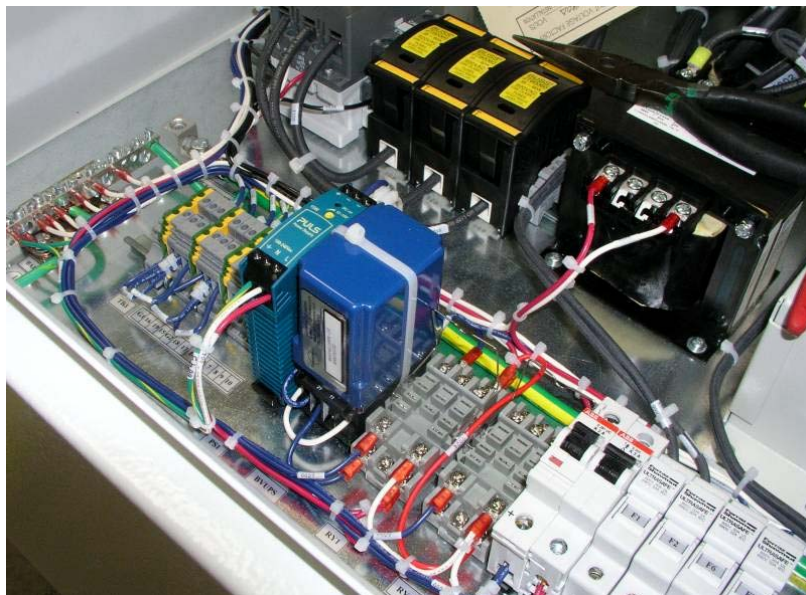
Reference Designator: BVUPS

Tools: Wire snips

1. Open front door on panel
2. Cut the tie-wraps securing the UPS to the DIN rail
3. Remove the old UPS
4. Remove the UPS base from the DIN rail by pushing to one side and lifting off the rail
5. Install new tie-wraps onto the base and reinstall onto the DIN rail



6. Install new UPS and secure using tie-wrap



6- ENVIRONMENTAL CONDITIONS FOR TRANSPORT AND STORAGE

Ambient Temperature Range:
Non operating, packaged: -40°C to +85°C

Relative Humidity Range:
10 – 90% non-condensing

Atmospheric Pressure Range:
Sea level to 10,000 feet

7- CONTACT INFORMATION

Office Locations:

Main Office:

TEAL Electronics

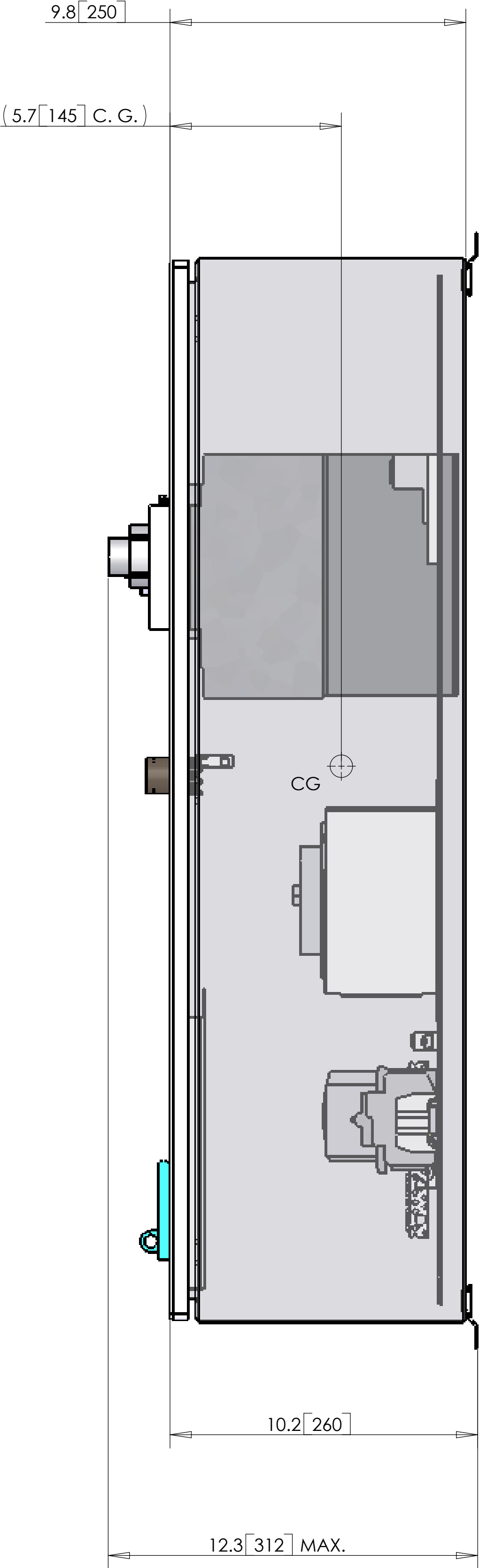
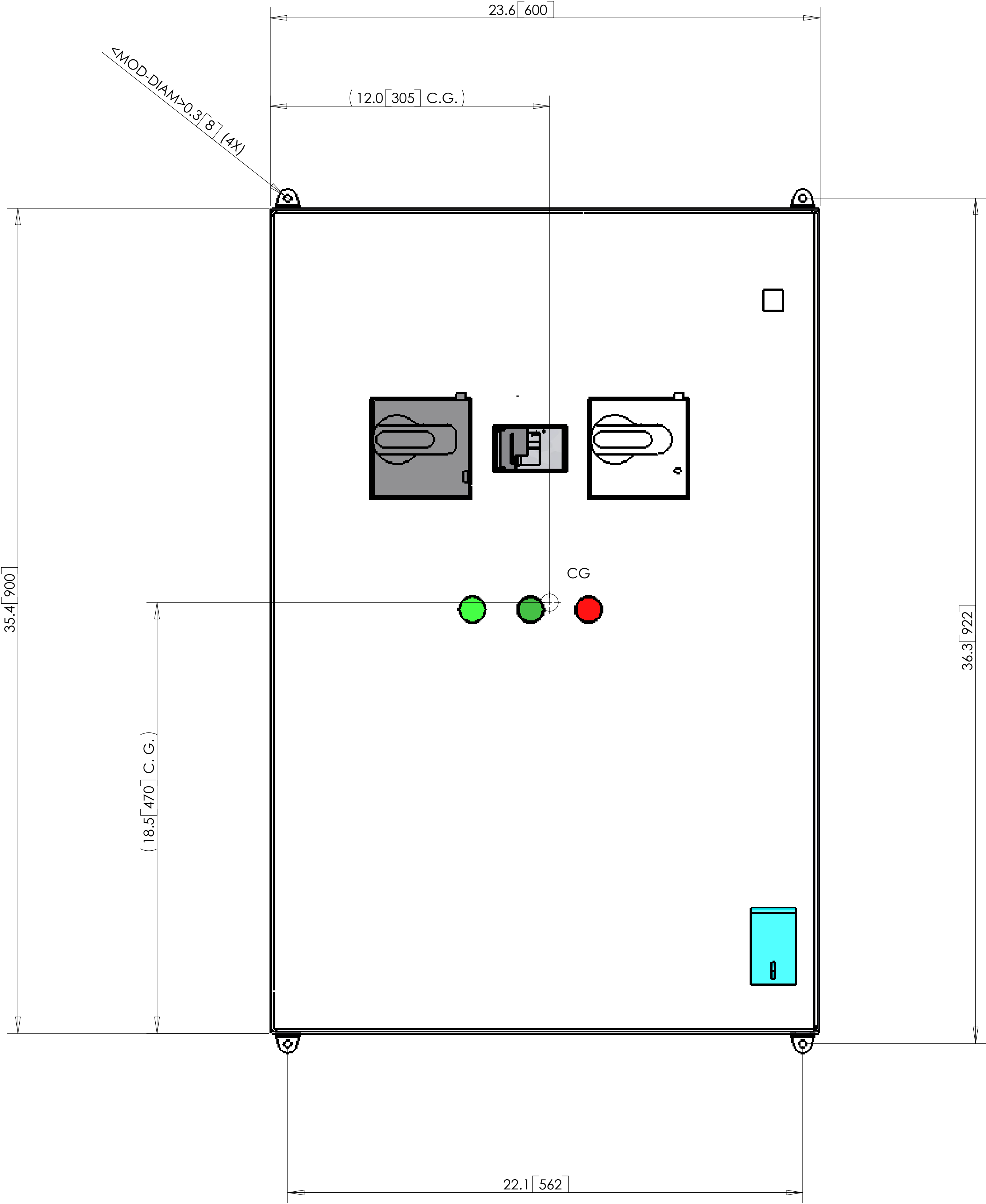
10350 Sorrento Valley Road
 San Diego, CA 92121
 (800) 888-TEAL or (800) 888-8325
 (858) 558-9000
 (858) 558-9500 FAX

Office Hours: 8AM to 5PM PST

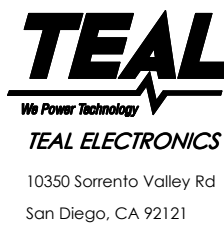
8- REVISION HISTORY

Revision	A0	Prototype Release, ECO 5267	
Check By:		Date:	04-16-09
Approve By:		Date:	04-17-09
Revision	A1	Production Release, ECO 5425	
Approve By:		Date:	06-12-09
Revision	A2	As per ECR 09223, changed current rating to match specifications.	
Approve By:		Date:	10-21-09
Revision	A3	As per ECR 09317, added FRU replacement procedures and added specifications and schematic of the UPS	
Approve By:		Date:	1-07-10

Revisions			
REV.	DESCRIPTION	DATE/DRAWN BY	APPROVED BY
X0	PROTOTYPE RELEASE. ECO 5267	3/11/09	EBG
A0	PRODUCTION RELEASE. ECO 5425	6/9/2009	HZ



NOTES:

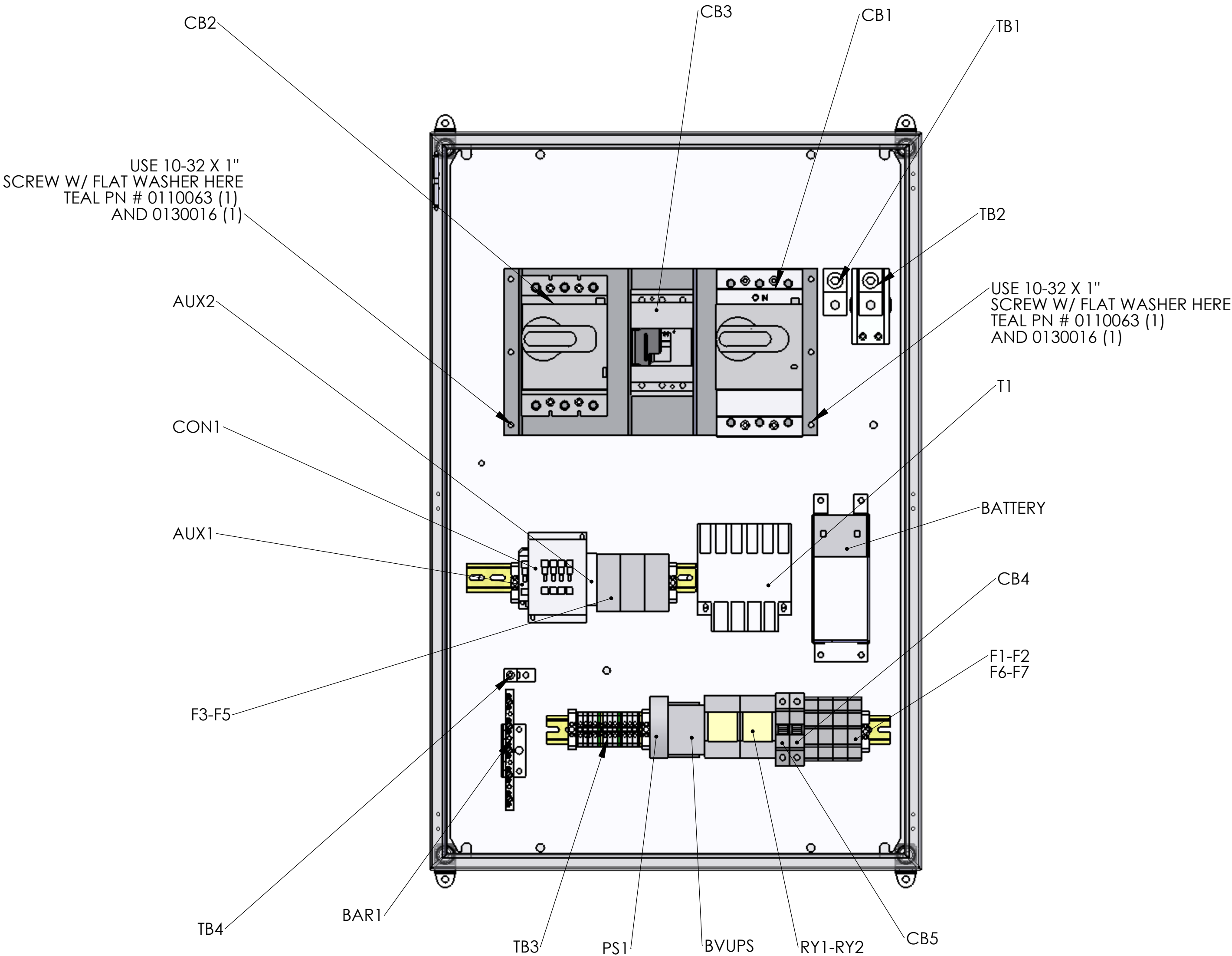
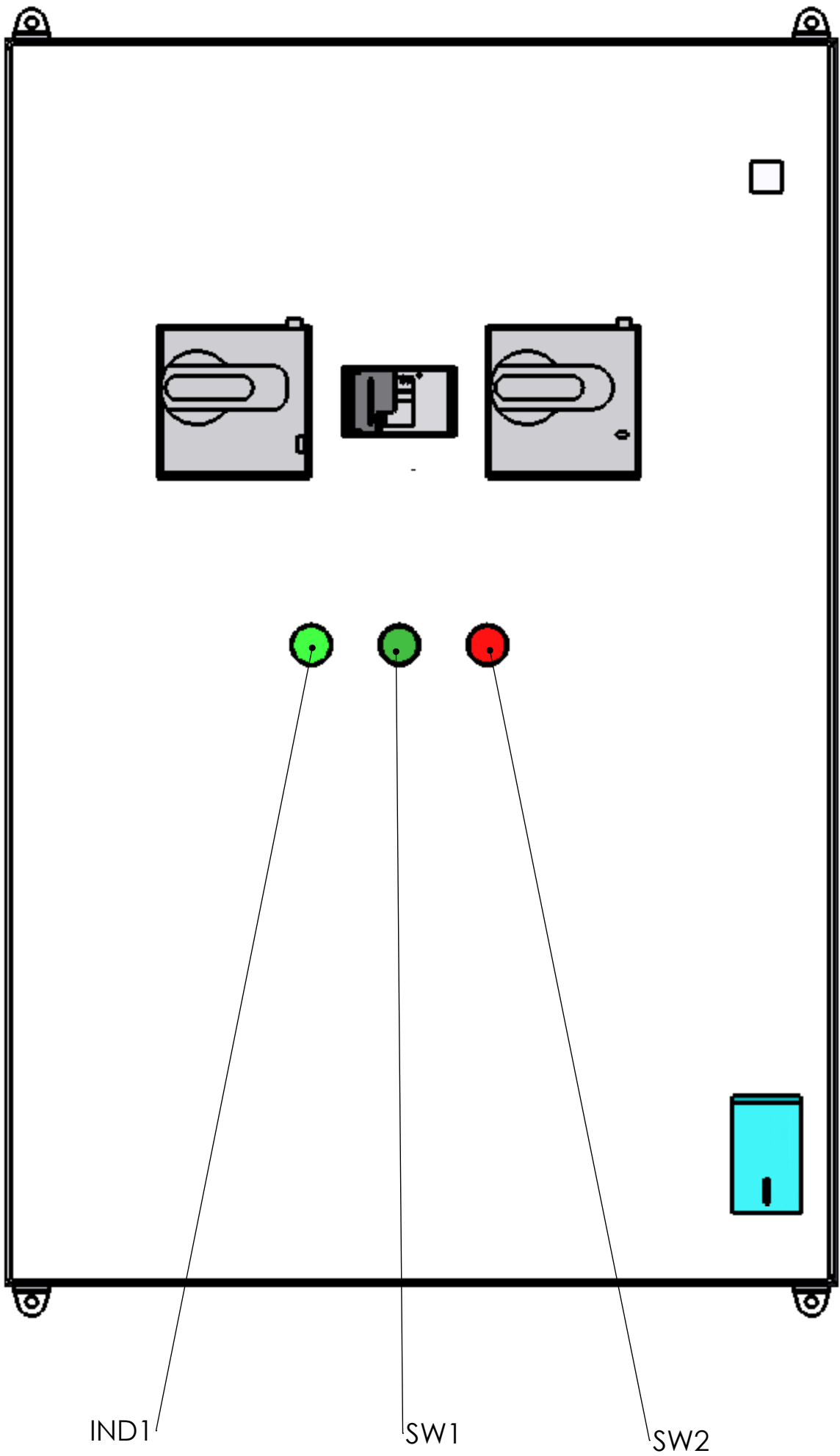


THIS DOCUMENT CONTAINS CONFIDENTIAL PROPRIETARY INFORMATION AND IS NOT TO BE DISTRIBUTED IN WHOLE OR IN PART WITHOUT THE WRITTEN PERMISSION OF TEAL ELECTRONICS CORPORATION.

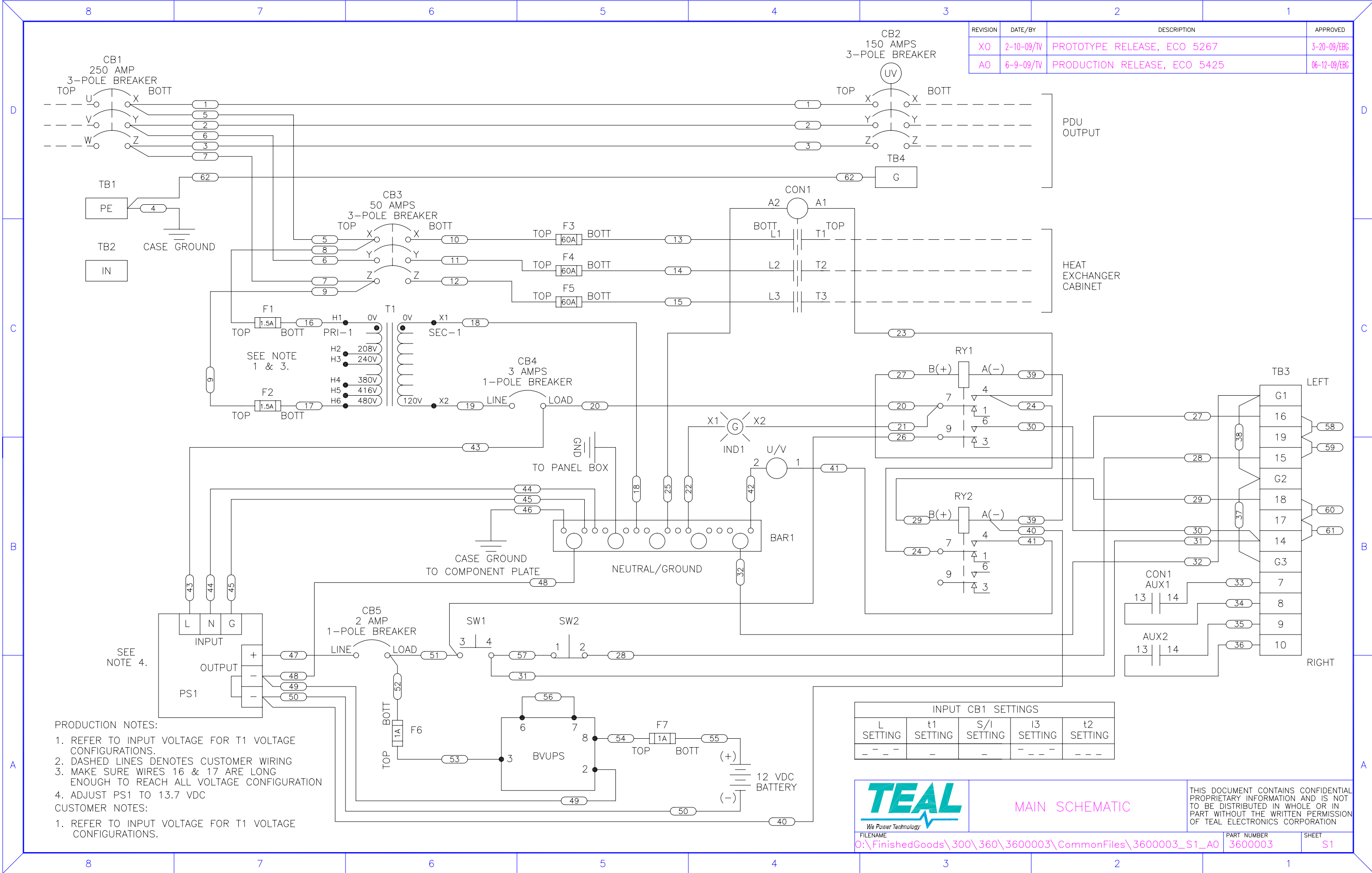
DIMENSION DIAGRAM

FILENAME	PART NUMBER	SHEET
O:\FinishedGoods\300\360\3600003\CommonFiles\3600003_D1_A0	3600003	D1

Revisions			
REV.	DESCRIPTION	DATE/DRAWN BY	APPROVED BY
X0	PROTOTYPE RELEASE. ECO 5267	3-18-09	EBG
A0	PRODUCTION RELEASE. ECO 5425	6/9/2009	HZ



NOTES:



REVISION	DATE/BY	DESCRIPTION	APPROVED
X0	2-10-09/IV	PROTOTYPE RELEASE, ECO 5267	3-20-09/EBG
A0	6-9-09/IV	PRODUCTION RELEASE, ECO 5425	06-12-09/EBG

INPUT CB1 SETTINGS				
L SETTING	t1 SETTING	S/I SETTING	I3 SETTING	t2 SETTING
- - -	-	-	- - -	- - -



MAIN SCHEMATIC

THIS DOCUMENT CONTAINS CONFIDENTIAL PROPRIETARY INFORMATION AND IS NOT TO BE DISTRIBUTED IN WHOLE OR IN PART WITHOUT THE WRITTEN PERMISSION OF TEAL ELECTRONICS CORPORATION

FILENAME O:\FinishedGoods\300\360\3600003\CommonFiles\3600003_S1_A0	PART NUMBER 3600003	SHEET S1
--	------------------------	-------------

- PRODUCTION NOTES:
1. REFER TO INPUT VOLTAGE FOR T1 VOLTAGE CONFIGURATIONS.
 2. DASHED LINES DENOTES CUSTOMER WIRING
 3. MAKE SURE WIRES 16 & 17 ARE LONG ENOUGH TO REACH ALL VOLTAGE CONFIGURATION
 4. ADJUST PS1 TO 13.7 VDC
- CUSTOMER NOTES:
1. REFER TO INPUT VOLTAGE FOR T1 VOLTAGE CONFIGURATIONS.



[Top](#) » [Catalog](#) » [DC UPS System](#) » [BVUPS](#)

[My Account](#) | [Cart Contents](#) | [Checkout](#)

Shopping Cart



BVUPS12PFA-10

\$79.95

0 items

Categories

- ▾ [Toshiba PLCs](#) (261)
- ▾ [Panel Meters](#) (48)
- ▾ [Test Equipment](#) (5)
- ▾ [EPROM Programmer](#) (115)
- ▾ [EPROM Emulator](#) (7)
- ▾ [Power Supplies](#) (28)
- ▾ [DC UPS System](#) (7)
- ▾ [Optivisor](#) (26)
- ▾ [Humidity-Temp logger](#) (4)
- ▾ [Tower-Lighting Alarm Monitor](#) (4)
- ▾ [Floobydust](#) (82)

Newest Products

Go to..

DC UPS System

Search



Use keywords to find the product you are looking for.

[Advanced Search](#)
[ShoppingCart Map](#)

Information

[Shipping & Returns](#)
[Privacy Notice](#)
[Conditions of Use](#)
[Contact Us](#)

What's New?



[EX10-MLK11](#)
\$579.00

Transtronics Battery Voltage UPS™ (BVUPS™) 12V version with power Fail contacts and special light load option

Reviews

Add to Cart



[Click to enlarge](#)

The Transtronics
BVUPS™
protects your
investment in
batteries and
increases system
reliability

- Are your lead-acid batteries prematurely dying a **deep discharge death**?
- Does it seem silly to have a 12 or 24 volt battery powered UPS powering a 12 or 24 volt power supply?
- Upon loss of DC power from your power supply, the **BVUPS™** "bumplessly" transfers your load to battery current. If the duration of the power failure drains the battery to its **Use EndVoltage(UEV)**, it will disconnect the load from the battery to protect the battery from damage due to deep discharge stress.
- Upon restoration of power, the **BVUPS™** recharges your batteries at a controlled rate with power supplied by your regulated power supply. It then maintains your batteries in a fully charged state.
- With the **Power Fail (PF)** option the **BVUPS™** with can sense the use of battery current to conditionally close a set of contacts indicating that the system is on battery power. As the battery runs down, the battery power is disconnected and the contacts remain closed (NC).

The BVUPS12PFA-10 replaces the BVUPS12PF-10

The BVUPS12PFA replaces the BVUPS12PF

Also see the [battery applications on line hand book](#) and [BVUPS FAQ](#)

See [Appropriate socket to use with the BVUPS](#) .

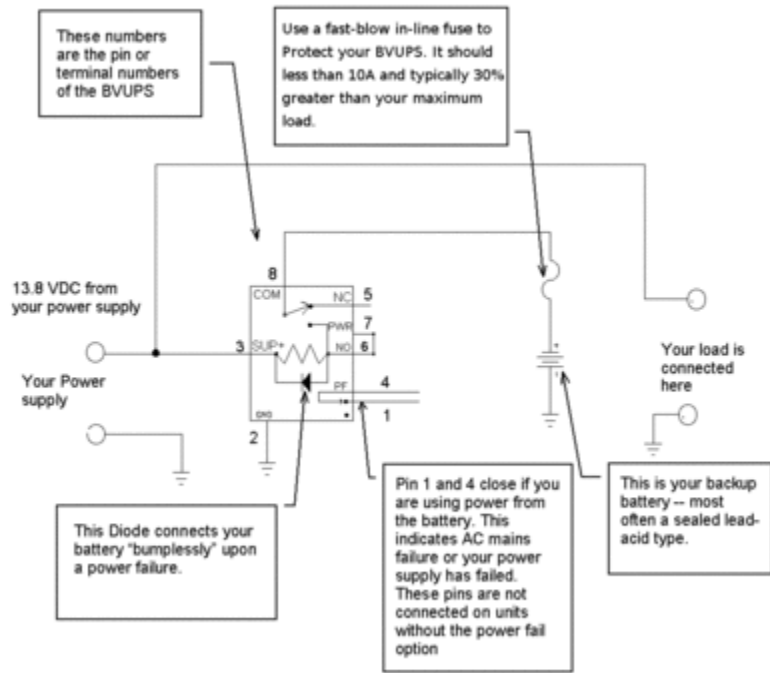
See [The full line of BVUPS units](#) .

See [Power supplies to use with the BVUPS](#)

See [Batteries to use with the BVUPS](#)

Specifications

Maximum ratings	
Input voltage	30 Vdc
Load current	6 amperes at 14 Vdc
Contact Ratings (battery and power fail)	Conservatively rated for 10 amps resistive at 240 Vac or 28 Vdc and 15 amps at 240 Vac or 28 Vdc
Dielectric strength	1000 Vac, 60 Hz for one second
Physical	
Ambient Operating Temperature	0 C to 60 C
Octal Case Size	2.75 X 1.75 X 2.375 inches (excluding octal pins)
Electrical	
Protection	Diode protected for reverse and over voltage conditions
Power consumption	500 milliwatts MAX 24 mA @ 13.6 Vdc 17 mA @ 10.0 Vdc
Supplied voltage	10-14 (20-28) VDC (your equipment must be able to operate in these ranges. Most 12 volt equipment is designed to work from 9 to 15 V)
Drop out threshold	10 V +/-200 mV (20V +/-500 mV)
Drop out delay	10 seconds of continuous low voltage
Initial charge rate	1.5 A (Can be programmed to higher values with external charge rate module)
Power fail minimum current	400 mA (Special -10 125mA versions available)



[Click to enlarge](#)

For more information, see this [page](#).

[Reviews](#)

[Add to Cart](#)

Product Link [BVUPS12PFA-10](#)

index 32968-0.35

Customers who bought this product also purchased

[BAT12V7AH](#)

[OCT-SOC](#)

Monday 09 February 02:02:12 PM, 2009

Transtronics, Inc.
3209 W.9th street
Lawrence, KS 66049
USA

Ph [+1\(785\) 841 3089](tel:+17858413089)
FAX [+1\(785\) 841 0434](tel:+17858410434)
Email inform@xtronics.com
WEB <http://xtronics.com>
Cart map <http://secure.transtronics.com>

12-27-10



SSL CERTIFICATE

[Corporate Home Page](#)

(C) Copyright 1994-2009, Transtronics, Inc. All rights reserved
Transtronics® is a registered trademark of Transtronics, Inc.
All trademarks are the property of their respective owners.



MATERIAL SAFETY DATA SHEET

PS, PSH, PSG, PG and Power Sport Series Valve Regulated (VRLA) Batteries Absorbed Electrolyte (AGM)

Section 1 - Product Identification

Manufacturers Name Power-Sonic Corporation, 7550 Panasonic Way San Diego, CA 92154	Emergency Telephone Numbers: CHEMTREC (Domestic): (800) 424-9300 CHEMTREC (International): (703) 527-3887
	Telephone Number for Information Power-Sonic Corporation: (619) 661-2020
	Date Issued: November 19, 2008

The information contained within is provided as a service to our customers and is for their information only. The information and recommendations set forth herein are made in good faith and are believed to be accurate at the date compiled. Power-Sonic Corporation makes no warranty expressed or implied.

Section 2 - Hazardous Ingredients/Identity Information

Components	CAS Number	Approx Wt. %	OSHA PEL (µg/m ³)	ACGIH TLV (µg/m ³)	NIOSH (µg/m ³)
Inorganic Lead/Lead Compounds	7439-92-1	65%-75%	50	150	10
Tin	7440-31-5	<0.5%	2000	2000	N/A
Calcium	7440-70-2	<0.1%	N/A	N/A	N/A
Electrolyte: Dilute sulfuric Acid	7664-93-9	14-20%	1000	1000	1000
Fiberglass Separator	-	5%	N/A	N/A	N/A
Case Material: Acrylonitrile Butadine Styrene (ABS)	9003-56-9	5-10%	N/A	N/A	N/A

Inorganic lead and electrolyte (sulfuric acid) are the main components of every Valve Regulated Lead Acid battery supplied by Power-Sonic Corporation. Other ingredients may be present dependent upon the specific battery type. For additional information contact Power-Sonic Corporation Technical Department.

Section 3 - Physical/Chemical Characteristics

Components	Density	Melting Points	Solubility (H ₂ O)	Odor	Appearance
Lead	11.34	621 F°	None	None	Silver-Gray
Lead Sulfate	6.20	1950 F°	40mg/l(60 F°)	None	White Powder
Lead Dioxide	9.40	554 F°	None	None	Brown Powder
Sulfuric Acid	About 1.30	203-240 F°	100%	Sharp penetrating pungent	Clear Colorless Liquid
Fiberglass Separator	N/A	N/A	Slight	None	White Fibrous
Case Material: Acrylonitrile Butadine Styrene (ABS)	N/A	N/A	None	None	Solid

Section 4 – Flammability Data

Components	Flashpoint	Explosive Limit	Comments
Lead and Sulfuric Acid	None	None	None
Hydrogen		LEL = 4.1%	Sealed batteries can emit hydrogen if overcharged (float voltage > 2.40 VPC)
Fiberglass Separator	N/A	N/A	Toxic vapors may be released. In case of fire, wear self contained breathing apparatus
Acrylonitrile Butadiene Styrene (ABS)	None	N/A	Temp over 527°F (300°C) may release combustible gases. In case of fire, wear self contained breathing apparatus

Section 5 - Reactivity Data

Stability	Unstable		Conditions to Avoid
	Stable	X	Prolonged overcharge on high current, ignition sources. Sulfuric acid remains stable at all temperatures
Incompatibility (Materials to Avoid) Sulfuric acid: Contact with combustibles and organic materials may cause fire and explosion. Also reacts violently with strong reducing agents, metals, sulfur trioxide gas, strong oxidizers, and water. Contact with metals may produce toxic sulfur dioxide fumes and may release flammable hydrogen gas. Lead Compounds: Avoid contact with strong acids, bases, halides, halogenates, potassium nitrate, permanganate, peroxides, nascent hydrogen, and reducing agents.			
Hazardous Decomposition or Byproducts Sulfuric acid: Sulfur trioxide, carbon monoxide, sulfuric acid mist, sulfur dioxide, and hydrogen sulfide. Lead Compounds: High temperatures above the melting point are likely to produce toxic metal fume, vapor, or dust; contact with strong acid or base or presence of nascent hydrogen may generate highly toxic arsine gas. Hazardous Polymerization.			
Polymerization: Sulfuric acid will not polymerize			
Decomposition Products: Sulfuric Dioxide, Trioxide, Hydrogen Sulfide, Hydrogen.			
Conditions to Avoid: Prohibit smoking, sparks, etc. from battery charging area. Avoid mixing acid with other chemicals.			

Section 6 - Health Hazard Data

Routes of Entry Sulfuric acid: Harmful by all routes of entry Lead compounds: Hazardous Exposure can occur only when product is heated, oxidized, or otherwise processed or damaged to create dust, vapor or fume.
Inhalation Sulfuric Acid: Breathing sulfuric acid vapors and mists may cause severe respiratory problems. Lead Compounds: Dust or fumes may cause irritation of upper respiratory tract or lungs. Fiberglass Separator: Fiberglass is an irritant to the upper respiratory tract, skin and eyes. For exposure up to 10F use MSA Comfoll with type H filter. Above 10F use Ultra Twin with type H filter. This product is not considered carcinogenic by NTP or OSHA.
Skin Contact Sulfuric acid: Severe irritation, burns and ulceration. Lead Compounds: Not absorbed through the skin

Continued on next page

Ingestion

Sulfuric acid: May cause severe irritation of the mouth, throat, esophagus, and stomach.

Lead Compounds: May cause abdominal pain, nausea, vomiting, diarrhea, and severe cramping. Acute ingestion should be treated by a physician.

Eye Contact

Sulfuric acid: Severe irritation, burns, cornea damage and possible blindness.

Lead Compounds: May cause eye irritation.

Acute Health Hazards

Sulfuric acid: Severe skin irritation, burns, damage to cornea may cause blindness, upper respiratory irritation.

Lead Compounds: May cause abdominal pain, nausea, headaches, vomiting, loss of appetite, severe cramping, muscular aches and weakness, and difficulty sleeping. The toxic effects of lead are cumulative and slow to appear. It affects the kidneys, reproductive and central nervous systems. The symptoms of lead overexposure are listed above. Exposure to lead from a battery most often occurs during lead reclamation operations through the breathing or ingestion of lead dust or fumes.

Chronic Health Hazards

Sulfuric acid: Possible scarring of the cornea, inflammation of the nose, throat and bronchial tubes, possible erosion of tooth enamel.

Lead Compounds: May cause anemia, damage to kidneys and nervous system, and damage to reproductive system in both males and females.

Carcinogenicity

Sulfuric acid: The National Toxicological Program (NTP) and The International Agency for Research on Cancer (IARC) have classified strong inorganic acid mist containing sulfuric acid as a Category 1 carcinogen, a substance that is carcinogenic to humans. The ACGIH has classified strong inorganic acid mist containing sulfuric acid as an A2 carcinogen (suspected human carcinogen). These classifications do not apply to liquid forms of sulfuric acid or sulfuric acid solutions contained within a battery. Inorganic acid mist (sulfuric acid mist) is not generated under normal use of this product. Misuse of the product, such as overcharging, may result in the generation of sulfuric acid mist.

Lead Compounds: Human studies are inconclusive regarding lead exposure and an increased cancer risk. The EPA and the International Agency for Research on Cancer (IARC) have categorized lead and inorganic lead compounds as a B2 classification (probable/possible human carcinogen) based on sufficient animal evidence and inadequate human evidence.

Medical Conditions Generally Aggravated by Exposure

Inorganic lead and its compounds can aggravate chronic forms of kidney, liver, and neurological diseases. Contact of battery electrolyte (acid) with the skin may aggravate skin diseases such as eczema and contact dermatitis. Overexposure to sulfuric acid mist may cause lung damage and aggravate pulmonary conditions.

Emergency and First Aid ProceduresInhalation

Sulfuric acid: Remove to fresh air immediately. If breathing is difficult, give oxygen

Lead Compounds: Remove from exposure, gargle, wash nose and lips, consult physician

Ingestion

Sulfuric acid: Do not induce vomiting, consult a physician immediately.

Lead Compounds: Consult a physician immediately

Eyes

Sulfuric acid: Flush immediately with water for 15 minutes, consult a physician.

Lead Compounds: Flush immediately with water for 15 minutes, consult a physician

Skin

Sulfuric acid: Flush with large amounts of water for at least 15 minutes, remove any contaminated clothing. If irritation develops seek medical attention.

Lead Compounds: Wash with soap and water.

Section 7 - Precautions for Safe Handling and Use

Steps to be Taken in Case Material is Released or Spilled

There is no release of material unless the case is damaged or battery is misused/overcharged. If release occurs stop flow of material, contain/absorb all spills with dry sand, earth, or vermiculite. Do not use combustible materials. Neutralize spilled material with soda ash, sodium bicarbonate, lime, etc. Wear acid-resistant clothing, boots, gloves, and face shield. Dispose of as hazardous waste. Do not discharge acid to sewer

Waste Disposal Method

Spent Batteries - send to secondary lead smelter for recycling. Follow applicable federal, state and local regulations. Neutralize as in preceding step. Collect neutralized material in sealed container and handle as hazardous waste as applicable. A copy of this MSDS must be supplied to any scrap dealer or secondary lead smelter with the battery.

Precautions to be Taken in Handling and Storing

Store batteries in a cool, dry, well ventilated area that are separated from incompatible materials and any activities which may generate flames, sparks, or heat. Keep all metallic articles that could contact the negative and positive terminals on a battery and create a short circuit condition.

Electrical Safety

Due to the battery's low internal resistance and high power density, high levels of short circuit current can be developed across the battery terminals. Do not rest tools or cables on the battery. Use insulated tools only. Follow all installation instructions and diagrams when installing or maintaining battery systems.

Fiberglass Separator

Fiberglass is an irritant to the upper respiratory tract, skin and eyes. For exposure up to 10F°/ use MSA Comfoll with type H filter. Above 10F use Ultra Twin with type H filter. This product is not considered carcinogenic by NTP or OSHA.

Section 8 - Control Measures

Respiratory Protection

None required under normal conditions. If battery is overcharged and concentrations of sulfuric acid are known to exceed PEL use NIOSH or MSH approved respiratory protection.

Engineering Controls

Store and handle batteries in a well ventilated area. If mechanical ventilation is used, components must be acid resistant

Protective Gloves

None needed under normal conditions. If battery case is damaged use rubber or plastic elbow length gauntlets

Eye Protection

None needed under normal conditions. If handling damaged or broken batteries use chemical splash goggles or face shield

Other Protective Clothing or Equipment

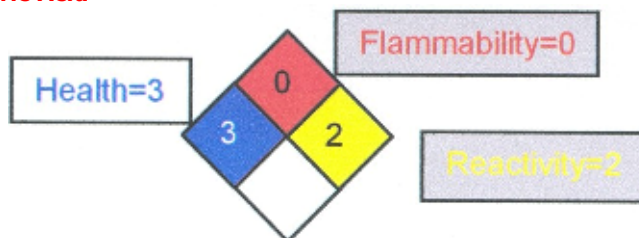
None needed under normal conditions. In case of damaged or broken battery use an acid resistant apron. Under severe exposure or emergency conditions wear acid resistant clothing.

Work Hygienic Practices

Handle batteries carefully to avoid damaging the case. Do not allow metallic articles to contact the battery terminals during handling. Avoid contact with the internal components of the battery.

Section 9 Regulatory Information

NFPA Hazard Rating for Sulfuric Acid



Transportation Batteries. Non-Restricted Status

North America Surface and Air Shipments

Our nonspillable lead acid batteries are listed in the U.S. Department of Transportation's (DOT) hazardous materials regulations but are **excepted** from these regulations since they meet all of the following requirements found at 49 CFR 173.159(d) – NMFC # 60680 Class 65.

- When offered for transport, the batteries are protected against short circuits and securely packaged as required by 49 CFR 173.159(d) (1);
- The batteries and outer packaging are marked with the words NONSPILLABLE BATTERY as required by 49 CFR 173.159(d) (2); and
- The batteries comply with the vibration and pressure differential tests found in 49 CFR 173.159(d) (3) and "crack test" found at 49 CFR 173.159(d) (4).

International

Our non-spillable lead acid batteries also are **excepted** from the international hazardous materials (also known as "dangerous goods") regulations since they comply with the following requirements:

- The vibration and pressure differential tests found in Packing Instruction 806 and Special Provision A67 of the **International Air Transport Association (IATA) Dangerous Goods Regulations**;

The vibration and pressure differential tests found in Packing Instruction 806 and Special Provision A67 of the International Civil Aviation Organization (ICAO) Technical Instructions for the Safe Transport of Dangerous Goods by Air; and

- The vibration, pressure differential, and "crack" tests found in Special Provision 238.1 and 238.2 of the **International Maritime Dangerous Goods (IMDG) Code**.

Regulatory Information

RCRA: Spent lead acid batteries are not regulated as hazardous waste by the EPA when recycled, however state and international regulations may vary.

CERCLA (superfund) and EPCRA:

- Reportable Quantity (RQ) for spilled 100% sulfuric acid under CERCLA (superfund) and EPCRA (Emergency Planning Community Right to Know Act) is 1,000lbs. State and local reportable quantities for spilled sulfuric acid may vary.
- Sulfuric acid is a listed "Extremely Hazardous Substance" under EPCRA with a Threshold Planning Quantity (TPQ) of 1,000lbs.
- EPCRA Section 302 Notification is required if 1,000lbs. or more of sulfuric acid is present at one site. The quantity of sulfuric acid will vary by battery type. Contact Power-Sonic Corporation for additional information.
- EPCRA Section 312 Tier 2 reporting is required for batteries if sulfuric acid is present in quantities of 500lbs. or more and/or lead is present in quantities of 10,00lbs. or more.
- Supplier Notification: This product contains toxic chemicals which may be reportable under EPCRA Section 313 Toxic Chemical Release Inventory (Form R) requirements. If you are a manufacturing facility under SIC codes 20 through 39 the following information is provided to enable you to complete the required reports:

Regulatory Information continued:

(f)

Toxic Chemical	CAS Number	Approximate % by weight
Lead	7439-92-1	60
Sulfuric Acid	7664-93-9 10-	30
Arsenic	7440-38-2	0.2

If you distribute this product to other manufacturers in SIC codes 20 through 39, this information must be provided with the first shipment in a calendar year. The Section 313 supplier notification requirement does not apply to batteries which are "consumer products". Not present in all battery types. Contact Power-Sonic Corporation for further information.

TSCA

Ingredients in Power-Sonic Corporation's batteries are listed in the TSCA Registry as follows:

Components	CAS Number	TSCA Status
Electrolyte Sulfuric Acid (H ₂ SO ₄)	7664-93-9	Listed
Inorganic Lead Compound: Lead (Pb)	7439-92-1	Listed
Lead Oxide (PbO)	1317-36-8	Listed
Lead Sulfate (PbSO ₄)	7446-14-2	Listed
Arsenic (As)	7440-38-2	Listed
Calcium (Ca)	7440-70-2	Listed
Tin (Sn)	7440-31-5	Listed

Power-Sonic Corporation
 7550 Panasonic Way,
 San Diego, CA 92154
 Tel: 619-661-2020
 Fax: 619-661-3650
 E-Mail: quality-assurance@power-sonic.com
 Website: <http://www.power-sonic.com>